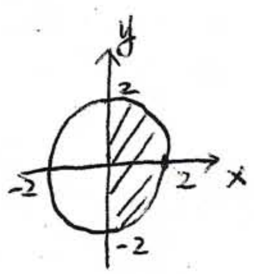
Math 241, F12 **Name:** Answer
Quiz 10 **SSN:** xxx-xx-

<u>Instructions:</u> There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No calculator is allowed in the quiz.

Problem 1. (5 points) Evaluate the iterated integral $\int_0^{\pi/4}\int_0^{4\cos\theta} r\,dr d\theta$.

$$\int_0^{\pi/4}\int_0^{4\cos\theta} r\,dr\,d\theta = \int_0^{\pi/4}\left[\frac{r^2}{2}\right]_{r=0}^{r=4\cos\theta} d\theta$$
$$= \int_0^{\pi/4} 8\cos^2\theta\,d\theta$$
$$= \int_0^{\pi/4} 4(1+\cos 2\theta)\,d\theta$$
$$= 4\left[\theta + \frac{1}{2}\sin 2\theta\right]_{\theta=0}^{\theta=\pi/4}$$
$$= 4\left[\frac{\pi}{4} + \frac{1}{2}\sin\frac{\pi}{2}\right]$$
$$= \pi + 2$$

Problem 2. (5 points) Compute the double integral $\iint_R \sqrt{4-x^2-y^2}\,dA$ where $R = \{(x,y) \mid x^2+y^2 \le 4, x \ge 0\}$ by changing to polar coordinates.



$$R = \left\{(r,\theta) \,\middle|\, -\frac{\pi}{2} \le \theta \le \frac{\pi}{2},\ 0 \le r \le 2\right\}$$
$$\sqrt{4-x^2-y^2} = \sqrt{4-(x^2+y^2)} = \sqrt{4-r^2}$$

Thus

$$\iint_R \sqrt{4-x^2-y^2}\,dA = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\int_0^2 \sqrt{4-r^2}\,r\,dr\,d\theta$$
$$= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left[-\frac{1}{3}(4-r^2)^{3/2}\right]_{r=0}^{r=2} d\theta$$
$$= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{8}{3}\,d\theta = \frac{8}{3}\left[\theta\right]_{\theta=-\pi/2}^{\theta=\pi/2}$$
$$= \frac{8}{3}\pi$$